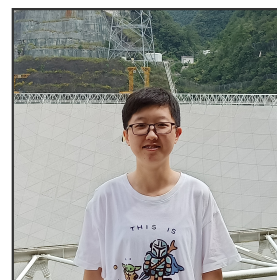


Chen Xu

PHD CANDIDATE · NAOC

National Astronomical Observatories, Chinese Academy of Sciences, Beijing, 100101, China

✉ xuchen@nao.cas.cn | 🏠 stellarxu.github.io | 📧 StellarXu | 📄 StellarXu | ORCID



Personal Information

Gender: **Female**

Nationality: **Chinese**

Date/Place of Birth: **1999.05**, Qinhuangdao, Hebei Province, China

Research Interest: **HI galaxy survey**, radio astronomy, Galaxy formation and evolution

Education

National Astronomical Observatories, Chinese Academy of Sciences (NAOC)

Beijing, China

PHD IN ASTROPHYSICS

2021.07 - present

- Advisor: Dr. Jie Wang
- Computational Cosmology Group, CCG

University of Chinese Academy of Sciences (UCAS)

Beijing, China

PHD IN ASTROPHYSICS

2021.07 - present

- School of Astronomy and Space Science

Jilin University (JLU)

Changchun, China

BS DEGREE IN PHYSICS

2017.09 - 2021.07

- School of Physics

Professional Experience

2026- **Server Administrator (Interim)**, NAOC, CCG

2023- **Key member of Hundred-deg² HI Deep (HD²) Survey Observational Group**, NAOC, CCG

2020- **Key member of HiFAST Technical Group**, NAOC, CCG

Publications

PUBLISHED

Chen Xu, Jie Wang, Yingjie Jing, et al. The FAST Hundred-Deg² HI Deep (HD²) Survey: Early Results from the Pilot Survey, *The Astrophysical Journal Supplement Series*, 2026, 285, 25. [Link]

Chen Xu, Jie Wang, Yingjie Jing, et al. HiFAST: An H I data calibration and imaging pipeline for FAST III. Standing Wave Removal, *Research in Astronomy and Astrophysics*, 2025, 25, 015011. [Link]

Yingjie Jing, Jie Wang, **Chen Xu**, et al. HiFAST: An H I data calibration and imaging pipeline for FAST, *SCIENCE CHINA Physics, Mechanics & Astronomy*, 2024, 67(5), 259514. [Link]

Zhang, C.P., Zhu, M., Jiang, P., et al. The FAST all sky H I survey (FASHI): The first release of catalog, *Science China Physics, Mechanics & Astronomy*, 2024, 67, 219511. [Link]

Ziming Liu, Jie Wang, Yingjie Jing, **Chen Xu**, et al. Observation of HI around three satellite galaxies of the M31 with the FAST: Andromeda II, NGC 205, and NGC 185, *Research in Astronomy and Astrophysics*, 2024. [Link]

Tiantian Liang, Jie Wang, Tao Wang, Yingjie Jing, **Chen Xu**, et al. Identification of Galactic H I Counterpart of the Large [O III] Emission Structure 1.2° Southeast of M31, *The Astronomical Journal*, 2025, 170(5), 253. [Link]

More publications can be found on my ADS library, click here.

IN PREP

Chen Xu, Jie Wang, Yingjie Jing, et al. Detection of the HI disk of M31 to a radius of 63 kpc, in preparation.

Awards, Fellowships, & Grants

2025	Zhihu New Knowledge Youth of 2024 (Top Honor for Zhihu Users), Zhihu	
2024	2nd Astronomy Popular Science Collection Activity, 1st Prize (Collaborated), Astronomy Popular Science Work Collaboration Group	
2024	Science Fiction and Popular Science Essay Contest, 1st prize for Popular Science and 3rd prize for sci-fiction, UCAS	RMB 5000 & 1000
2023	Science Fiction and Popular Science Essay Contest, Excellent Prize, UCAS	RMB 500
2022	1st UCAS Women's Football Tournament, Champion, UCAS	
2022	Three Good Student, UCAS	
2021	UCAS Graduate Scholarship, UCAS	RMB 1000
2018-2021	First Class Scholarship, Jilin University	RMB 2000/yr

Presentations

- 2026.01.15. *A Large Diffuse HI Disk of $N_{\text{HI}} = 10^{17-18} \text{ cm}^{-2}$ Rotating around M31 and Photoionization Analysis*. Student-organized HI Unofficial Discussion Club (LOC: Organizer).
- 2025.09.28. *The FAST Hundred-Deg2 HI Deep (HD2) Survey I. Early Results from the Pilot Survey*. 2nd FAST Scientific Meeting, Pingtang.
- 2025.05.25. *Pilot Survey of the FAST Hundred-Deg2 HI Deep (HD2) Survey I. Early Results from the Extragalactic HI Catalog*. Guo Shoujing Meeting, Changsha.
- 2024.11.19. *A Pilot Survey for the Hundred-deg2 HI Deep (HD2) Survey: Optical Counterparts and Scaling Relations with DESI*. Annual meeting of the FAST project in PKU/KIAA.
- 2023.10.26. *Low-z HI Extragalactic Surveys of FAST: Observations, Data Reduction and some Early Results*. Invited talk on USTC Journal Club (Online).
- 2021.11.23. *Introduction of Ripples Removal and RFI Flagging in HiFAST*. HiFAST Tech meeting in Beijing.

Professional Development

OBSERVING EXPERIENCE

- **FAST:** Hundreds of hours of observation planning and HI data reduction as a key member of FAST key projects (ZD2021_4 – ZD2025_4).
- **Optical:** Lijiang 2.4m CHILI IFU (1 night), Xinglong 2.16m (2 nights).
- **Interferometers:** Data reduction experience with VLA and GMRT (Co-I, 35 hours).

SKILLS

FAST Data Processing: Proficient in processing RAW data from various FAST observational modes, including HI point source catalog generation.

Simulation: Basic knowledge; mock HI galaxy from IllustrisTNG.

Languages: Mandarin, English, Python, Shell, ein bisschen Deutsch und un poco Español.

MEMBERSHIPS

Head of the HI Unofficial Discussion Small Group among students and postdocs.

Mentoring

2025- **Xu, Yiwei**, Master student, NAOC

Outreach

2024- **Outstanding Astronomy Contributor on Zhihu**, [click here to view](#)

2021-2023 **University of Chinese Academy of Sciences Astronomical Association**, Key member of the Science Popularization Department and Spring Equinox Project

2018-2019 **Jilin University Astronomy Society**, Head of the Academic Department